Stage Three – Configure Web/Database Access

To complete this portion of the project you will need to be able to accomplish the following tasks:

- Create a privileged MariaDB account so you can administer your database by logging in from your client
- Create a read-only MariaDB account that other students can use to log into your MariaDB server and read from it
- Configure your MariaDB Server for remote access
- Modify the instrument flat file (.csv file) and import it into your MariaDB database
- Configure your Apache Server to use the provided script as it's default web page
- Test your setup by accessing your database from your Web Browser (Local Host)
- Test your setup by accessing another student's database from your Web Browser (Remote Host)

It is recommended that you download all relevant files and scripts to your client and then SCP the various files to correct machines where applicable.

MariaDB Modifications

Creating a Read-Only User For Your MariaDB Server

This is an important step – **you don't want other students purposefully or accidentally making changes to your database.** However, other students will need to have the ability to log into your database to read its contents from their web browser. For our purposes **<u>ALL STUDENTS will create the following account:</u>**

User: csp450ro

Password: csp450ro

- Google the following:
  - “**how to create a Maria DB read only user**” to create the user
  - **“how to configure MariaDB for remote access”** to enable remote access to your MariaDB server
  - Test logging into the server **Locally** (from your Server machine) and **Remotely** (from your Client Machine.)

Still having trouble logging in? DID YOU RESTART THE MARIADB SERVICE?

Importing your Excel Flat File

The following procedure will guide you through creating your database and populating it with the flat file you edited:

- Edit the .csv file **populatedTable.csv** and add 10 new instruments, conditions and prices.
- Create a database called **inventory**
- Create a table in this database with the following commands (you can enter each one at the MariaDB command line one at a time):

```
CREATE TABLE instruments (

        id INT AUTO_INCREMENT PRIMARY KEY,

        instrument_type VARCHAR(100),

        instcondition VARCHAR(50),

        price DECIMAL(10,2)

);
```

- Use another terminal window to copy **(scp)** the csv file to **/var/lib/mysql/inventory/ on your SERVER machine**
- Login to your database again
- Import the flat file with the following commands you can enter each one at the MariaDB command line one at a time):

```
LOAD DATA INFILE 'populatedTable.csv' INTO TABLE inventory.instruments
```

```
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

- Execute the command **SELECT * FROM** instruments and confirm the records were imported

Configuring the Apache Server

Editing the PHP Script For Your Database Front End

- Download the file **phpScriptFrontEnd.txt** to your Client.
- Find the section in the script that you see below and change the username and password to match the read-only account you created in the previous section, and change the database name to **'inventory' (This was the name of the database we created in the previous section before we imported the flat file.)**

```
if (empty($errors)) {

// Database connection parameters

$username = 'ababamahmoudi';

 $password = '******';

$dbname = 'instruments';
```

- Once you have made the correct edits, save the file.

Copying Files To Your Server And Confirmation Of Package Installation

- Copy this file to your **Server** into the directory **/var/www/html** and **change the name to index.php**
- Remove the file **/var/www/html/index.html** from your **Server**

- Be sure that you have installed PHP, PHP MySQL plugin, and the MariaDB client:
    - **sudo apt update**
    - **sudo apt install apache2 php php-mysql mariadb-client**

- Lastly, restart the apache2 service:
    - **sudo systemctl restart apache2**

Final Testing

- From your Client, point your browser to the IP address of the Server. You should see the Instrument Lookup page. Using your Server IP as the Database Server IP, attempt to query one of the custom instruments you added to your file.
- If you know of another person who has their database up and running, enter their IP as the Database Server IP and attempt to query one of the custom instruments that they added to their file.

Visual Inspection

- From your Client, demonstrate that you can log into your MariaDB database using the read only user that you created as per the specification
- From your Client, demonstrate that you can log into another student's MariaDB database using the read only user that they created as per the specification
- Display your flat file and show the custom additions you made (your unique 10 instruments)
- From your client, access your database and perform a search for a New Guitar at a price of $1500
- From your client, access your database and perform a search that will display a unique instrument from your inventory
- From your client, access another student's database in the classroom and perform a search that will display a unique instrument from their inventory

Documentation Submission

Please submit the following to Black Board:

- The PHP script you are using as the front end of your Apache Server
- The instrument flat-file you imported into your MySQL database